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Trench closing assembly

Abstract

A trench closing assembly including at least one soil deflector.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of PCT Application No. PCT/IB2020/052424, filed Mar. 17, 2020, which claims the benefit of priority to U.S. Provisional Patent Application No. 62/831,338 filed Apr. 9, 2019, and U.S. Provisional Patent Application No. 62/967,293 filed Jan. 29, 2020. The entireties of all the foregoing listed applications are incorporated herein by reference.

BACKGROUND

(1) It is well known that good seed-to-soil contact within the seed trench is a critical factor in uniform seed emergence and high yields. To obtain good seed-to-soil contact, once seeds are planted in a seed trench, the seed trench needs to be closed so that soil surrounds the seed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side elevation view of an embodiment of a row unit of an agricultural planter.
- (2) FIG. 2 is a diagram of a system for implementing operational control of the closing wheel assembly based on signals generated by the trench closing sensor.
- (3) FIG. 3 is a left side perspective view of an embodiment of a trench closing assembly.
- (4) FIG. 4 is a top plan view of the trench closing assembly of FIG. 3.
- (5) FIG. 5 is a bottom plan view of the trench closing assembly of FIG. 3.
- (6) FIG. 6 is a top perspective view of the trench closing assembly of FIG. 3 with some parts

removed.

(7) FIG. 7 is a bottom perspective view of the trench closing assembly of FIG. 6.

(8) FIG. 8 is a bottom perspective view of the trench closing assembly of FIG. 7 with some parts removed.

(9) FIG. 9 is a bottom perspective view of the trench closing assembly of FIG. 8 with some parts removed.

(10) FIG. 10 is a top perspective view of the trench closing assembly of FIG. 9.

(11) FIG. 11 is a top perspective view of the frame of the trench closing assembly of FIG. 3.

(12) FIG. 12 is a bottom perspective view of the frame of FIG. 11.

(13) FIG. 13 is bottom plan view of the frame of FIG. 11.

(14) FIG. 14 is top perspective view of the frame of the trench closing assembly of FIG. 3.

(15) FIG. 15 is a bottom perspective view of the frame of FIG. 14.

(16) FIG. 16 is a top perspective view of the connection bracket of the frame of FIG. 14.

(17) FIG. 17 is a bottom perspective view of the connection bracket of FIG. 16.

(18) FIG. 18 is a top perspective view of the attachment bracket of the frame of FIG. 14.

(19) FIG. 19 is a bottom perspective view of the attachment bracket of FIG. 15.

(20) FIG. 20 is top perspective view of the mounting arm of the trench closing assembly of FIG. 3.

(21) FIG. 21 is a bottom perspective view of the mounting arm of FIG. 20.

(22) FIG. 22 is a front elevation view of the handle assembly of the trench closing assembly of FIG. 3.

(23) FIG. 23 is a left side elevation view of the handle assembly of FIG. 22.

(24) FIG. 24 is a perspective view of the shank of the handle assembly of FIG. 22.

(25) FIG. 25 is a bottom perspective view of the pressure sensor of the handle assembly of FIG. 22.

(26) FIG. 26 is a top perspective view of the pressure sensor of FIG. 25.

(27) FIG. 27 is a perspective view of the handle assembly and mounting arm of the trench closing assembly of FIG. 3.

(28) FIG. 28 is a perspective view of the trench closing assembly of FIG. 3 further including soil deflectors and a fluid applicator according to an embodiment.

(29) FIG. 29 is a left side elevation view of the trench closing assembly of FIG. 28.

(30) FIG. 30 is a top plan view of the trench closing assembly of FIG. 28.

(31) FIG. 31 is a left side elevation view of the closing wheel soil deflector of FIG. 28.

(32) FIG. 32 is a left side elevation view of the press wheel soil deflector of FIG. 28.

(33) FIG. 33 is a perspective view of the closing wheel soil deflector of FIG. 31.

(34) FIG. 34 is a perspective view of the press wheel soil deflector of FIG. 32.

(35) FIG. 35 is a top plan view of the closing wheel soil deflector of FIG. 31.

(36) FIG. 36 is a top plan view of the press wheel soil deflector of FIG. 32.

(37) FIG. 37 is a perspective view of the fluid applicator according to an embodiment.

(38) FIG. 38 is a right side elevation view of the fluid applicator of FIG. 37.

(39) FIG. 39 is a top plan view of the fluid applicator of FIG. 37.

(40) FIG. 40 is a rear elevation view of the fluid applicator of FIG. 37.

(41) FIG. 41 is a top plan view of a fluid applicator according to an embodiment.

(42) FIG. 42 is a left side elevation view of the closing wheel soil deflector of FIG. 31 and including a nozzle.

(43) FIG. 43 is a left side elevation view of the closing wheel soil deflector of FIG. 32 and including a nozzle.

(44) FIG. 44A is a perspective view of the trench closing assembly of FIG. 3 with a modified guard and further including a scraper with an optional fluid applicator according to an embodiment.

(45) FIG. 44B is a top plan view of the trench closing assembly of FIG. 44A.

(46) FIG. 44C is a bottom plan view of the trench closing assembly of FIG. 44A.

(47) FIG. 44D is a left side elevation view of the trench closing assembly of FIG. 44A.

- (48) FIG. 44E is a right side elevation view of the trench closing assembly of FIG. 44A.
 - (49) FIG. 45A is a top plan view of left and right scrapers of FIG. 44A.
 - (50) FIG. 45B is a bottom plan view of left and right scrapers of FIG. 45A.
 - (51) FIG. 45C is a front elevation view of left and right scrapers of FIG. 45A.
 - (52) FIG. 45D is a rear elevation view of left and right scrapers of FIG. 45A.
 - (53) FIG. 46A is a right side elevation view of a right scraper of FIG. 45A.
 - (54) FIG. 46B is a left side elevation view of a right scraper of FIG. 45A.
 - (55) FIG. 47A is a left side elevation view of a left scraper of FIG. 45A.
 - (56) FIG. 47B is a right side elevation view of a left scraper of FIG. 45A.
 - (57) FIG. 48A is a partial perspective view of the trench closing assembly of FIG. 44A with parts removed for viewing.
 - (58) FIG. 48B is a bottom plan view of the trench closing assembly of FIG. 48A.
 - (59) FIG. 48C is a right side elevation view of the trench closing assembly of FIG. 48A.
 - (60) FIG. 48D is a left side elevation view of the trench closing assembly of FIG. 48A.
 - (61) FIG. 49A is a perspective view of the guard of FIG. 44A.
 - (62) FIG. 49B is a top plan view of the guard of FIG. 49A.
 - (63) FIG. 49C is a bottom plan view of the guard of FIG. 49A.
 - (64) FIG. 49D is a front elevation view of the guard of FIG. 49A.
 - (65) FIG. 49E is a rear elevation view of the guard of FIG. 49A.
 - (66) FIG. 49F. is right side elevation view of the guard of FIG. 49A.
 - (67) FIG. 49G. is left side elevation view of the guard of FIG. 49A.
 - (68) FIG. 49H is a front elevation view of the guard of FIG. 49D with the adjustable plate removed.
- DESCRIPTION

(69) All references cited herein are hereby incorporated by reference in their entireties. In the event of a conflict in a definition in the present disclosure and that of a cited reference, the present disclosure controls.

(70) Referring now to the drawings, wherein like reference numerals designate identical or corresponding parts throughout the several views, FIG. 1 illustrates an embodiment of an agricultural planter row unit **200**. The row unit **200** is comprised of a frame **204** pivotally connected to a toolbar **202** by a parallel linkage **206** enabling each row unit **200** to move vertically independently of the toolbar **202**. The frame **204** operably supports one or more hoppers **208**, a seed meter **210**, a seed delivery mechanism **212**, a downforce control system **214**, a seed trench opening assembly **220**, a trench closing assembly **250**, an optional packer wheel assembly **260**, and an optional row cleaner assembly **270**. It should be understood that the row unit **200** shown in FIG. 1 may be for a conventional planter or the row unit **200** may be a central fill planter, in which case the hoppers **208** may be replaced with one or more mini-hoppers and the frame **204** modified accordingly as would be recognized by those of skill in the art.

(71) The optional downforce control system **214** is disposed to apply lift and/or downforce on the row unit **200** such as disclosed in U.S. Publication No. US2014/0090585. The downforce applied by downforce control system **214** can be determined by methods disclosed in U.S. Pat. No. 9,173,339.

(72) The seed trench opening assembly **220** includes a pair of opening discs **222** rotatably supported by a downwardly extending shank member **205** of the frame **204**. The opening discs **222** are arranged to diverge outwardly and rearwardly so as to open a v-shaped trench **10** in the soil **11** as the planter traverses the field. The seed delivery mechanism **212**, such as a seed tube or seed conveyor, is positioned between the opening discs **222** to deliver seed from the seed meter **210** into the opened seed trench **10**. The depth of the seed trench **10** is controlled by a pair of gauge wheels **224** positioned adjacent to the opening discs **222**. The gauge wheels **224** are rotatably supported by gauge wheel arms **226** which are pivotally secured at one end to the frame **204** about pivot pin **228**. A rocker arm **230** is pivotally supported on the frame **204** by a pivot pin **232**. It should be

appreciated that rotation of the rocker arm **230** about the pivot pin **232** sets the depth of the trench **10** by limiting the upward travel of the gauge wheel arms **226** (and thus the gauge wheels) relative to the opening discs **222**. The rocker arm **230** may be adjustably positioned via a linear actuator **234** mounted to the row unit frame **204** and pivotally coupled to an upper end of the rocker arm **230**. The linear actuator **234** may be controlled remotely or automatically actuated as disclosed, for example, in International Publication No. WO2014/186810.

(73) An optional downforce sensor **238** is configured to generate a signal related to the amount of force imposed by the gauge wheels **224** on the soil. In some embodiments the pivot pin **232** for the rocker arm **230** may comprise the downforce sensor **238**, such as the instrumented pins disclosed in U.S. Pat. No. 8,561,472.

(74) An optional seed meter **210** may be any commercially available seed meter, such as a finger-type meter or a vacuum seed meter. One exemplary vacuum seed meter is the VSet® meter, available from Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568.

(75) An optional packer wheel assembly **260** comprises an arm **262** pivotally attached to the row unit frame **204** and extends rearward of the closing wheel assembly **250** and in alignment therewith. The arm **262** rotatably supports a packer wheel **264**. An actuator **266** is pivotally attached at one end to the arm **262** and at its other end to the row unit frame **204** to vary the amount of downforce exerted by the packer wheel **264** to pack the soil over the seed trench **10**.

(76) An optional row cleaner assembly **270** may be the CleanSweep® system available from Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568. The row cleaner assembly **270** includes an arm **272** pivotally attached to the forward end of the row unit frame **204** and aligned with the trench opening assembly **220**. A pair of row cleaner wheels **274** are rotatably attached to the forward end of the arm **272**. An actuator **276** is pivotally attached at one end to the arm **272** and at its other end to the row unit frame **204** to adjust the downforce on the arm to vary the aggressiveness of the action of the row cleaning wheels **274** depending on the amount of crop residue and soil conditions.

(77) Referring to FIG. 2, a monitor **300** is visible to an operator within the cab of a tractor pulling the planter. The monitor **300** may be in signal communication with a GPS unit **310**, the trench closing assembly actuator **256** and the optional packer wheel assembly actuator **266** to enable operational control of the trench closing assembly **250** and the optional packer wheel assembly **260** based on the signals generated by trench closing sensors **1000**, which are described in International Publication No. WO2017/197274. Also as discussed later, the monitor **300** may be programmed to display operational recommendations based on the signals generated by the trench closing sensors **1000**. The monitor **300** may also be in signal communication with the row cleaner actuator **276**, the downforce control system **214**, the depth adjustment actuator **234** to enable operational control of row cleaner assembly **270**, the downforce control system **214** and the trench opening assembly **230**, respectively.

(78) Trench Closing Assembly

(79) Examples of Trench Closing Assemblies can be found in U.S. Patent Application Nos. 62/637,372, filed 1 Mar. 2018, 62/644,201, filed 16 Mar. 2018, 62/731,813, filed 14 Sep. 2018, 62/791,203, filed on 11 Jan. 2019, and in International Application No. PCT/US2019/020452, filed on 2 Mar. 2019. Also, an example of a trench closing assembly is the FurrowForce™ trench closing assembly available from Precision Planting, LLC.

(80) FIGS. 3-27 illustrate a trench closing assembly **250D** according to an embodiment. In this embodiment, trench closing assembly **250D** has a main frame **251D** that is connected to row unit **200**. Trench closing assembly **250D** has a frame **1520**, an actuator **259**, a pair of closing wheels **254D-1**, **254D-2**, and optionally, a press wheel **255D**.

(81) Turning to FIGS. 14-19, frame **251D** has a connection bracket **1540** and an attachment bracket **1550**. Connection bracket **1540** can have one or more bolts **1209D** extending through apertures **1207D** in connection bracket **1540** for mounting the frame **251D** to the row unit **200**. Connection

bracket **1540** (as with connection brackets **1200** and **1201A**) can be varied to mate attachment to different styles of row units. Connection bracket **1540** has a first side **1542-1**, a second side **1542-2**, and a plate **1543** disposed between first side **1542-1** and second side **1542-2**. First side **1542-1** and second side **1542-2** each have a post **1541-1** and **1541-2**, respectively, protruding perpendicularly outward.

(82) Attachment bracket **1550** connects to connection bracket **1540**. Attachment bracket **1550** provides a common structure for mounting other parts while connection bracket **1540** has a varied structure to mate with different styles of row units. Attachment bracket has a first side **1552-1**, a second side **1552-2**, a crossbar **1557** disposed between first side **1552-1** and second side **1552-2**, and plate **1553** disposed between first side **1552-1** and second side **1552-2**, and plate **1553**. First side **1552-1** and second side **1552-2** each have a u-shaped opening **1555-1** and

1555-2, respectively, for connection to posts **1541-1** and **1541-2**, respectively. First side **1552-1** and second side **1552-2** have openings **1558-1** and **1558-2**, respectively, for accepting pivots **1529-1** and **1529-2**, respectively. Attachment bracket **1550** can be secured to connection bracket with fastener **1549**. Optionally, attachment bracket **1550** can also have an opening **1501** disposed in first side **1552-1** or second side **1552-2** for accepting a pin. While connection bracket **1540** is illustrated with separate parts it may be fabricated as a unitary part.

(84) Optionally, as illustrated in FIGS. 5 and 7, a guard **1590** can be disposed on frame **1520** ahead of closing wheels **254D-1**, **254D-2**. Guard **1590** can prevent rocks, rootballs, or other trash from approaching closing wheels **254D-1**, **254D-2**. The height of guard **1590** can be adjusted by changing the placement of arms **1591-1** and **1591-2** on frame **1520**.

(85) Turning to FIGS. 20-21, mounting arm **1570** has a first side **1571-1**, a second side **1571-2**, a plate **1574** disposed between first side **1571-1** and second side **1571-2**, a hole **1573** through plate **1574** for passage of sleeve **1603**, axle hubs **1575-1** and **1575-2** for mounting press wheels **255D-1** and **255D-2**, respectively, and, optionally, a handle cradle **1579**.

(86) Turning to FIGS. 6-13, frame **1520** is pivotally connected to frame **251D** through pivots **1529-1** and **1529-2**. Frame **1520** has a first side **1526-1** and a second side **1526-2**, a plate **1525** connecting first side **1526-1** and second side **1526-2**, a connection arm **1527** extending rearwardly along the direction of travel from plate **1525**, a cross-brace **1528-1** connecting connection arm **1527** to first arm **1526-1**, a cross-brace **1528-2** connecting connection arm **1527** to second arm **1526-2**. First side **1526-1** and second side **1526-2** have openings **1521-1** and **1521-2** for disposing about pivots **1529-1** and **1529-2**, respectively. Mounting arm **1570** can attach to frame **1520** at connections **1522-1a** and **1522-2a** or to **1522-1b** and **1522-2b**. The plurality of connections allows the distance between closing wheels **254D-1**, **254D-2**, and press wheel **255D** (**255D-1**, **255D-2**) to be changed. There can be one connection **1522** or a plurality of connections **1522**. As the mounting arm **1570** position is changed, there are also corresponding connections **1523-a** and **1523-b** (matching in number to connection **1522**) for connecting handle assembly **1600** to frame **1520**. Frame **1520** also has connections **1524** (**1524-1a**, **1524-1b**, **1524-2a**, and **1524-2b**) for mounting closing wheels **254D-1**, **254D-2**. While there can be one set of connections **1524**, the plurality of connections **1524** allow for forward and back placement of the closing wheels **254D-1**, **254D-2** on frame **1520**, or closing wheels **254D-1**, **254D-2** can be offset from each other with one closing wheels **254D-1**, **254D-2** being mounted to a forward location (the “a” position) or to a rear location (the “b” position). As illustrated, closing wheels **254D-1**, **254D-2** are offset from each other. Optionally, frame **1520** can have an opening **1502** in either the first side **1526-1** or second side **1526-2** (shown in **1526-2**) for accepting a pin. Frame **1520** can be raised to allow opening **1501** and opening **1502** to align for accepting a pin (not shown). This allows trench closing assembly **250D** to be raised for transport or when closing is not needed. While illustrated on one side, openings **1501** and **1502** can be disposed on both sides.

(87) Actuator **259** is disposed between plate **1525** and plate **1553** to apply a force to plate **1553** to cause frame **1520** to pivot and apply pressure to closing wheels **254D-1** and **254D-2**.

(88) Turning to FIGS. 22-27, handle assembly **1600** is illustrated. Handle assembly **1600** has a sleeve **1603** having a first diameter **1620** and a second diameter **1621**. Second diameter **1621** is small enough to be disposed through load sensor **1610**, and first diameter **1620** is large enough so that it cannot pass through load sensor **1610**. Sleeve **1603** has a bracket **1622** (u-shaped bracket as illustrated or any other shape) for accepting handle **1609**. Disposed on sleeve **1603** below bracket **1622** are bevel washers **1601** and **1602**. Bevel washer **1601** and bevel washer **1602** are disposed with their concave surfaces facing each other. This allows bevel washers **1601** and **1602** to flex to absorb shocks experienced by trench closing assembly **250D** to prevent overloading of load sensor **1610**. Before bevel washers **1601** and **1602** reach maximum flex, first diameter **1620** will contact plate **1604** to limit the travel. Load sensor **1610** is a pancake load sensor. Load sensor **1610** has a hole **1612** for passage of sleeve **1603**. Disposed on the underside of load sensor **1610** are a plurality of feet **1611** to allow load sensor **1610** to flex and measure force. Load sensor **1610** can be disposed on mounting arm **1570** directly, or as shown, a plate **1604** can be disposed between load sensor **1610** and mounting arm **1570**. Optionally, a washer **1607** can be disposed about sleeve **1603** under mounting arm **1570**. Handle assembly **1600** is connected to frame **1520** at connection **1523-a** or **1523-b** with a bracket **1606** (u-shaped bracket) and a bolt **1605** connecting bracket **1606** with sleeve **1603**. In the horizontal position, handle **1609** locks handle assembly **1600** in place against mounting arm **1570**. In the vertical position, handle **1609** releases handle assembly from engagement with mounting arm **1570**. Bolt **1605** can be adjusted to set a vertical placement of mounting arm **1570** relative to frame **1520**.

(89) Load sensor **1610** can be connected to a network directly through a plug (not shown) having a CAN processor to allow direct communication over a CAN network. The CAN processor can communicate pressure readings and provide control signals over the CAN network. Alternatively, load sensor **1610** can be connected to a control module (either an on-row module, or a module controlling a plurality of row) to communicate pressure readings that are then processed by the control module.

(90) Soil Deflector

(91) FIGS. 28-30 illustrate soil deflectors (**3100**, **3500**) and fluid applicator (**2000**) disposed on closing assembly **250D**. Soil deflector **3100** is illustrated on the left, and the right (not shown) is the mirror image of the left. Soil deflectors **3100** and soil deflectors **3500** are illustrated as two parts, but they can be made unitary parts. Soil deflectors **3100** and **3500** can keep dirt/mud from spraying or “rooster tailing” as the trench closing system **250D** traverses the field. Also, soil deflectors **3100** and **3500** can scrape closing wheels **254D** and/or press wheels **255D**, respectively, by being adjusted to contact closing wheels **254D** and/or press wheels **255D**.

(92) Soil deflector **3100-2** is connected to axle **257-2** and extends rearward opposite the direction of travel adjacent to closing wheel **254D-2**. Depending on the shape of closing wheel **254D**, soil deflector **3100** can be shaped to follow the contour of closing wheel **254D** or contact closing wheel **254D**. Soil deflector **3100-2** can be made of two parts, bracket **3101-2** and deflector **3102-2**. When connected to axle **257-2**, optionally a spacer (not shown), such as washer or bushing, can be disposed on axle **257-2** to vary the spacing between soil deflector **3100-2** and closing wheel **254D-2**. Optionally, a connection orientation between bracket **3102-2** and deflector **3102-2** can be changed to change the orientation of deflector **3102-2** with closing wheel **254D-2**. In one embodiment, soil deflector **3100** extends beyond the circumference of closing wheel **254D**, and optionally, extends behind closing wheel **254D** toward frame **1520**. Deflector **3102-2** can be made from a flexible material. A flexible material is a material that will deflect when impacted by soil when trench closing assembly **250D** is operated at typical planting speeds (e.g., up to 20 mph (32 kph) or up to 10 mph (16 kph)). In one embodiment, deflector **3102-2** is polyurethane.

(93) Optionally, soil deflector **3500** (**3500-1**, **3500-2**) can be included for press wheels **255D** (**255D-1**, **255D-2**). While illustrated with two press wheels **255D** (**255D-1**, **255D-2**), there can be just one press wheel **255D**. Soil deflector **3500** (**3500-1**, **3500-2**) is connected to axle **258** (**258-1**,

258-2) and extends rearward opposite the direction of travel adjacent to press wheel **255D** (**255D-1**, **255D-2**). Depending on the shape of press wheel **255D**, soil deflector **3500** can be shaped to follow the contour of press wheel **255D** or press wheel **254D**. Soil deflector **3500-2** can be made of two parts, bracket **3501-2** and deflector **3502-2**. When connected to axle **258-2**, optionally a spacer (not shown), such as washer or bushing, can be disposed on axle **258-2** to vary the spacing between soil deflector **3500-2** and closing wheel **2545-2**. Optionally, a connection orientation between bracket **3502-2** and deflector **3502-2** can be changed to change the orientation of deflector **3502-2** with press wheel **255D-2**. In one embodiment, soil deflector **3500** extend beyond the circumference of press wheel **255D**, and optionally, extend behind press wheel **255D**. Deflector **3502-2** can be made from a flexible material as defined above.

(94) Fluid Applicator

(95) Fluid applicator **2000** is illustrated in FIGS. **37-40** according to one embodiment. Fluid applicator **2000** has a frame **2100** for connection to mounting arm **1570**. An optional slot **2101** can be disposed in frame **2100** to allow for adjustment front to back along the direction of travel. While illustrated with two arms **2104** (**2104-1**, **2104-2**), fluid applicator **2000** can have just one arm **2014**. Arms **2104** (**2104-1**, **2104-2**) are disposed on either side of mounting arm **1570**. Nozzles **2120** (**2120-1**, **2120-2**) are attached to arms **2104** (**2104-1**, **2104-2**), respectively. Optionally, arms **2104** (**2104-1**, **2104-2**) can have slots **2105** (**2105-1-a**, **2105-1-b**, **2105-2-a**, **2105-2-b**) to allow nozzles **2120** (**2120-1**, **2120-2**) to be adjusted vertically. For adjustable connection, U-bolts **2110** (**2110-1**, **2110-2**) can connect nozzle **2120** (**2120-1**, **2120-2**) to arms **2104** (**2104-1**, **2104-2**) through slots **2105** (**2105-1-a**, **2105-1-b**, **2105-2-a**, **2105-2-b**). Optionally, slots **2105** (**2105-1-a**, **2105-1-b**, **2105-2-a**, **2105-2-b**) can be sized larger than U-bolts **2110** (**2110-1**, **2110-2**) to allow nozzles **2120** (**2120-1**, **2120-2**) to rotate in a direction transverse to the direction of travel so that nozzles **2120** (**2120-1**, **2120-2**) can deposit fluid closer to or further away from the trench. Optionally, frames **2111** (**2111-1**, **2111-2**) can be included to ensure fit of nozzles **2120** (**2120-1**, **2120-2**) to arms **2104** (**2104-1**, **2104-2**), for example when nozzles **2120** (**2120-1**, **2120-2**) are round. Nozzles **2120** (**2120-1**, **2120-2**) are connected to a fluid source (not shown) for applying fluid to the soil surface. Optionally, frame **2100** can be disposed on mounting arm **1570** rotated 180° from the orientation illustrated.

(96) In an alternative embodiment, fluid applicator **2000'** is illustrated in FIG. **41**. While illustrated on the right side, fluid applicator **2000'** can also be installed as a mirror image on the left side. An arm **2200-1** is connected to an axle **257-1** between frame **1520** and closing wheel **254D-1** and extends in a rearward direction away from the direction of travel. A nozzle **2220-1** is attached to arm **2200-1** via a bracket **2210-1**. Optionally, a slot **2205-1** can be disposed in arm **2200-1** to permit adjustment of nozzle **2220-1** front to back along the direction of travel. Nozzle **2220-1** is connected to a fluid source (not shown) for applying fluid to the soil surface.

(97) In an alternative embodiment as illustrated in FIG. **42** or FIG. **43**, a nozzle **2300** (**2300-2**) can be connected to one or both soil deflectors **3100**, **3500**. Nozzle **2300** is connected to a fluid source (not shown). While illustrated with nozzle **2300** depositing fluid rearward of soil deflector **3100** or **3500** opposite the direction of travel, nozzle **2300** can be positioned to apply fluid towards the trench, or nozzle **2300** can be positioned to deposit fluid away from the trench.

(98) FIGS. **44A-44E** illustrate trench closing assembly **250E**, which is similar to trench closing assembly **250D** except that guard **1590** is replaced with guard **1590A**, and press wheel **255D** (**255D-1**, **255D-2**) is shown with press wheel **1300'** (**1300'-1**, **1300'-2**) from U.S. Provisional Patent Application No. 62/843,037.

(99) Scraper **4000** (**4000-1**, **4000-2**) is illustrated in FIGS. **45A** to **47B**. Scraper **4000** (**4000-1**, **4000-2**) connects to axel **257** (**257-1**, **257-2**, respectively) and is shaped to extend towards closing wheel **254D** (**254D-1**, **254D-2**, respectively) and follows the contour of closing wheel **254D** rearward opposite the direction of travel of closing system **250E** and can terminate at an edge of closing wheel **254D** or extend beyond the outer radius of closing wheel **254D**. Optionally, a nozzle **4010** (**4010-1**, **4010-2**) can be connected to scraper **4000** to apply fluid from a source (not shown).

Scraper **4000** (**4000-1**, **4000-2**) can be a unitary piece, or it can be fabricated from different pieces. Scraper **4000** (**4000-1**, **4000-2**) can have different sections. Hub section **4001** (**4001-1**, **4001-2**) is for connecting to axle **257** (**257-1**, **257-2**, respectively). Section **4002** (**4002-1**, **4002-2**) transitions from hub section **4001** (**4001-1**, **4001-2**) to scraping section **4003** (**4003-1**, **4003-2**, respectively). Scraping section **4003** (**4003-1**, **4003-2**) follow the contour of closing wheel **254D** (**254D-1**, **254D-2**, respectively). Scraping section **4003** (**4003-1**, **4003-2**) terminates at end **4004** (**4004-1**, **4004-2**, respectively). Optional nozzle **4010** (**4010-1**, **4010-2**) is connected to end **4004** (**4004-1**, **4004-2**, respectively). While illustrated with this embodiment, scraper **4000** can be adapted to work with any closing wheel **254** (such as **254**, **254D**).

(100) Guard **1590A** is illustrated in FIGS. **48A** to **49H**. Guard **1590** is generally U-shaped having a front plate **1593** and arms **1591-1** and **1591-2** extending rearwardly opposite the direction of travel. Front plate **1593** and arms **1591-1**, **1591-2** can be a unitary part or fabricated from individual parts. Front plate **1593** can be perpendicular to the direction of travel of closing system **250**, or front plate **1593** can be disposed at an angle with respect to the direction of travel. Front plate **1593** has a slot **1594** for accepting a fastener (such as a bolt or screw) to adjustably attach adjustable plate **1595**. Slot **1594** can be continuous (not shown) or can be a series of holes for specific positions for accepting the fastener. While shown with adjustable plate **1595**, guard **1590** can replace the front plate **1593** with slot **1594** with plate **1595** in place of front plate **1593** for a fixed configuration (not shown). Arm **1591-1** can have one hole **1597-1** or a plurality of holes **1597-1** (**1597-1a**, **1597-1b**, **1597-1c**). One hole **1597-1** (such as **1597-1b**) can be used for attaching guard **1590A** to frame **1520** and closing wheel **254D-1**. Closing wheel **254D-1** could optionally be attached at hole **1597-1a** or **1597-1c** for different configurations. Arm **1591-2** can have one hole **1597-2** or a plurality of holes **1597-2** (**1597-2b**, **1597-2c**) and/or **1598**. One hole **1597-2** (such as **1597-2b**) can be used for attaching guard **1590A** to frame **1520** and closing wheel **254D-2**. Closing wheel **254D-2** could optionally be attached at hole **1597-2c** or at hole **1598**. Attaching closing wheel **254D-2** at hole **1598** allows for an offset of closing wheels **254D-1**, **254D-2**. In this configuration, the angle created by front plate **1593** is such that the side of **1593** closest to closing wheel **254D-2** is disposed forward of the side of **1593** closest to closing wheel **254D-1**. Arms **1592-1** and **1592-2** extend from arms **1591-1** and **1591-2**, respectively, to connect guard **1590A** to frame **1520** at connections **1596-1** and **1596-2**, respectively. Arms **1591-1**, **1591-2**, **1592-1**, **1592-2** can be designed to follow the contour of frame **1520**. While illustrated as a separate part, guard **1590A** could be formed as part of frame **1520**.

(101) Various embodiments of the invention have been described above for purposes of illustrating the details thereof and to enable one of ordinary skill in the art to make and use the invention. The details and features of the disclosed embodiments are not intended to be limiting, as many variations and modifications will be readily apparent to those of skill in the art.

Claims

1. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from the main frame; a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; a first closing wheel soil deflector connected to an axle of the first closing wheel and extending opposite to the forward direction of travel beside the first closing wheel, wherein the soil deflector has a structure to keep soil from spraying as the trench closing

system traverses the field; wherein the first closing wheel soil deflector extends a distance beyond a diameter of the first closing wheel and extends toward the frame member behind the first closing wheel.

2. The trench closing system of claim 1, further comprising a second closing wheel soil deflector connected to an axle of the second closing wheel and extending opposite to the forward direction of travel beside the second closing wheel.

3. The trench closing system of claim 2, wherein the second closing wheel soil deflector extends a distance beyond a diameter of the second closing wheel and extends toward the frame member behind the second closing wheel.

4. The trench closing system of claim 2, wherein the second closing wheel soil deflector comprises a second bracket and a second deflector, and wherein the second deflector is a flexible material.

5. The trench closing system of claim 1, wherein the first closing wheel soil deflector comprises a first bracket and a first deflector, and wherein the first deflector is a flexible material.

6. The trench closing system of claim 1, further including a mounting arm pivotally mounted to the frame member, the mounting arm supporting at least one press wheel rearward of the closing wheels, and a first press wheel soil deflector connected to an axle of the at least one press wheel and extending opposite to the forward direction of travel beside the at least one press wheel.

7. The trench closing system of claim 6, wherein the first press wheel soil deflector extends toward a centerline of the trench closing system behind the at least one press wheel.

8. The trench closing system of claim 6, wherein the at least one press wheel comprises a first press wheel and a second press wheel with the first press wheel soil deflector connected to an axle of the first press wheel and further including a second press wheel soil deflector connected to an axle of the second press wheel.

9. The trench closing system of claim 8, wherein the second press wheel soil deflector comprises a first bracket and a first deflector, and wherein the first deflector is a flexible material.

10. The trench closing system of claim 6, wherein the first press wheel soil deflector further comprises a nozzle.

11. The trench closing system of claim 1, wherein the first closing wheel soil deflector further comprise a nozzle.

12. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from the main frame; a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; a first closing wheel soil deflector connected to an axle of the first closing wheel and extending opposite to the forward direction of travel beside the first closing wheel, wherein the soil deflector has a structure to keep soil from spraying as the trench closing system traverses the field; and a mounting arm pivotally mounted to the frame member, the mounting arm supporting at least one press wheel rearward of the closing wheels, and a first press wheel soil deflector connected to an axle of the at least one press wheel and extending opposite to the forward direction of travel beside the at least one press wheel; wherein the first press wheel soil deflector comprises a first bracket and a first deflector, and wherein the first deflector is a flexible material.

13. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from the

main frame; a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; a first closing wheel soil deflector connected to an axle of the first closing wheel and extending opposite to the forward direction of travel beside the first closing wheel, wherein the soil deflector has a structure to keep soil from spraying as the trench closing system traverses the field; and a mounting arm pivotally mounted to the frame member, the mounting arm supporting at least one press wheel rearward of the closing wheels, and a first press wheel soil deflector connected to an axle of the at least one press wheel and extending opposite to the forward direction of travel beside the at least one press wheel; and a fluid applicator disposed on the mounting arm to apply fluid to the soil surface.

14. The trench closing system of claim 13, wherein the fluid applicator comprises a frame that is disposed over the mounting arm and having a first arm on the first side of the seed trench and a second arm on the second side of the seed trench, wherein a first nozzle is attached to the first arm and a second nozzle is attached to the second arm.

15. The trench closing system of claim 14, wherein the fluid applicator has a slot for adjusting a position of the fluid applicator front to back along the direction of travel.

16. The trench closing system of claim 14, wherein the first arm has a slot for adjusting a vertical position of the first nozzle and the second arm has a slot for adjusting a vertical position of the second nozzle.

17. The trench closing system of claim 14, wherein the nozzle is adjustable transverse to the direction of travel.

18. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from the main frame; a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; a first closing wheel soil deflector connected to an axle of the first closing wheel and extending opposite to the forward direction of travel beside the first closing wheel, wherein the soil deflector has a structure to keep soil from spraying as the trench closing system traverses the field; and a first arm connected to the axle of the first closing wheel and disposed between the closing wheel and the frame member, the first arm extending opposite to the forward direction of travel and having a first nozzle for applying a fluid to the soil surface.

19. The trench closing system of claim 18, wherein the first arm has a slot for adjusting a position of the first nozzle front to back along the direction of travel.
